Pratik Adhikari

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Research Interests

Renewable Energy Integration, Power System Modeling and Dynamics, Distribution Systems, Power System Monitoring, and Data-Driven Grid Modernization

Education

Institute of Engineering, Pulchowk Campus Apr 2022 – Aug 2026
Bachelor of Engineering (BE), Electrical Engineering — Tribhuvan University, Lalitpur, Nepal

- **Aggregate: 84.72%** — Distinction (institutional scale; 80% distinction threshold).
- Final-year project: *Digital Twin for Real-Time Monitoring and Short-Term Forecasting of a Grid-Connected Photovoltaic System*; Supervisor: Asst. Prof. Deependra Neupane.
- Coursework: Power System Analysis I & II, Power Electronics, Engineering Mathematics, Control Systems, Switchgear & Protection, Instrumentation I & II, Utilization of Electrical Energy.

SOS Hermann Gmeiner Secondary School Gandaki 2019 – 2021
Higher Secondary (Grade 11–12), Science — GPA: 3.64 / 4.00

Honors & Awards

- **Ranked 3rd** in the graduating Electrical Engineering cohort, Institute of Engineering, Tribhuvan University (84.72% aggregate).
- **Merit-based undergraduate scholarship**, Institute of Engineering — awarded on the national engineering entrance examination (approx. 10,000 candidates) and received a separate **IOE Scholarship Stipend** for all eight semesters.

Technical Skills

Modeling & Simulation: MATLAB/Simulink, COMSOL Multiphysics, Proteus
Programming: Python, MATLAB, C/C++, Javascript
Power Systems & Electrical Design: DIgSILENT PowerFactory (coursework), AutoCAD
Data & Machine Learning: NumPy, Pandas, scikit-learn, statsmodels, Matplotlib
Instrumentation & Embedded Systems: Arduino, sensor calibration, serial data acquisition
Other: LaTeX, Git, Bluebeam Revu, PCB design

Research Experience

Digital Twin for Real-Time Monitoring & Short-Term Forecasting of a Grid-Connected PV System 2025 – 2026
Final-year project (four-member team), Department of Electrical Engineering, Pulchowk Campus; Supervisor: Asst. Prof. Deependra Neupane

- Built and parameterised a MATLAB/Simulink model of a 335 W grid-connected PV system and evaluated it against 41 paired measurements collected on site over two days, obtaining 34.76 W RMSE (24.1% nRMSE, $r = 0.918$) when driven by on-site irradiance against 114.23 W RMSE ($r = 0.703$) when driven by satellite-derived irradiance, quantifying the benefit of site-specific measurement.
- Designed and calibrated a low-cost Arduino/LDR irradiance sensor against a co-located smartphone-based reference (power-law fit, $R^2 = 0.974$, $n = 21$), and used it with temperature sensing to drive the model from live serial input, demonstrating continuous sensor-to-simulation operation.
- Developed a SARIMA$(1,1,1) \times (1,1,1)_{24}$ pipeline for day-ahead irradiance and temperature forecasting ($R^2 = 0.921$ and 0.964) and propagated the forecasts through the model, predicting next-day PV energy within 0.5% of the value obtained under measured driving conditions.

- Analysed irradiance-dependent residuals, attributing low-irradiance underprediction to sensor floor limitations and high-irradiance overprediction to unmodelled thermal and loss effects, and documented both as targets for instrument and model refinement.

Deadbeat Predictive Current Control with SVPWM and LCL Filtering for Grid-Connected Voltage-Source Converters 2026 – Present
Ongoing research study, Department of Electrical Engineering, Pulchowk Campus

- Studying deadbeat predictive current control with space-vector PWM for a three-phase grid-connected VSC: discrete-time dq-frame modelling, fixed-frequency 10 kHz modulation, and capacitor-current-feedback active damping of an LCL grid interface.
- Developing MATLAB/Simulink simulations of a 5 kW, 400 V, 50 Hz system comparing L and LCL filtering through resonance, harmonic-spectrum, and P–Q dynamic-response analyses; preliminary results indicate markedly lower current THD for the deadbeat-SVPWM with LCL active-damping configuration at matched average switching frequency.
- Examining LCL resonance, digital-delay effects on active damping, and controller de-tuning as practical limits on closed-loop stability; the study is currently being extended toward a revised control formulation.

Professional Experience

DeltaQ Engineering — Electrical Engineer (Remote, MEP) May 2026 – Present
US-based MEP engineering consultancy

- Perform electrical load calculations and system analysis for residential, multi-family, commercial, and industrial projects, documenting in accordance with applicable US codes and standards.
- Produce technical single-line diagrams, specifications, and coordination layouts in AutoCAD, and evaluate electrical equipment selections against project and code requirements.

Academic Projects

Power Plant Design 2026

- Designed a 16 MW run-of-river hydropower plant end-to-end: site selection, hydrological analysis, turbine selection, generator/transformer specification, neutral grounding, station supply, diesel-generator backup, and battery-bank sizing.

Industrial Power Distribution & Illumination 2025

- Designed the electrical distribution system for a multi-storey commercial building in AutoCAD—lighting/power-socket layouts, SDB/FDB/MDB sizing, cable selection, protective-device coordination, and 250 kVA transformer/330 A MCCB selection.

Transmission & Distribution System Design 2026

- Designed a 132 kV, double-circuit transmission line for a 108 MW, 220 km corridor including voltage-level selection, tower geometry, insulation design, and conductor evaluations.

Smart Fertigation System 2023

- Built an Arduino-based monitoring and control system (DHT-11, soil moisture sensors) automating irrigation with prototype fertilisation control.

Teaching, Mentoring & Outreach

BloomED — STEM Content Creator (Volunteer) Jun 2021 – Jan 2022

- Produced simplified learning material explaining technical and scientific concepts for underprivileged students, developing science-communication and teaching skills.

Languages

Nepali (native); English (professional working proficiency); Hindi (conversational)